

ConvertAudioEngine-Keep71.ps1 v3.1.2

Retains TrueHD 7.1 and AAC 7.1 audio tracks

Keep original TrueHD 7.1 / AAC 7.1 and add a **DDP 5.1 (EAC3) compatibility track** (Pass+Copy)

- Downmix all other 7.1 audio to **DDP 5.1**
- Re-encode 5.1 audio to **DDP 5.1**
- DTS-HD MA/HRA re-encodes at **DDP 5.1**
- Remove 2-channel commentary tracks
- Per-language winner selection (PreTrack-QRS): keep the highest-priority track.

1. To Adjust Bitrate (line 71-76)

```
You can safely adjust these numbers, within '....k'
Common EAC3 5.1 bitrates: 384, 448, 512, 640, 768
Higher pipeline bitrates: 1024, 1152, 1280, 1408, 1536

$BitRateConfig = @{
    Downmix51      = '1152k'    # << Change your Bitrate
    ReEncode51     = '768k'     # << Change your Bitrate
    PassCopy51     = '1152k'    # << Change your Bitrate
    EncodeDTSHD    = '1024k'    # << Change your Bitrate
}
```

Remember to save the file after making changes.

All four values can be adjusted safely — changes apply automatically throughout the script.

Key	Default	Used for
Downmix51	1152k	All 7.1 sources downmixed to 5.1 (EAC3/DTS/PCM/FLAC)
ReEncode51	768k	5.1 sources re-encoded to DD+
PassCopy51	1152k	DDP 5.1 compatibility track added alongside TrueHD 7.1 / AAC 7.1
EncodeDTSHD	1024k	DTS-HD MA and HRA sources (5.1 and 7.1)

Requirements

PowerShell 7.6 - [Click here to Download PS 7.6 \(Windows x64 .msi\)](#)

FFmpeg 8.1 - <https://ffmpeg.org/download.html>

Quick Start (Windows 11)

1. Place **ConvertAudioEngine-Keep71.ps1**, **ffmpeg.exe**, and **ffprobe.exe** in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\ConvertAudioEngine-Keep71.ps1 ".\YourMovie.mkv"
```

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, **ffmpeg**, and **ffprobe** in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

3. Run:

```
pwsh -File ./ConvertAudioEngine-Keep71.ps1 "./YourMovie.mkv"
```

Output filename: The program saves a new file in the same folder, adding a codec suffix like **_keep71mix.mkv**. Your original file stays untouched.

Supported Audio Codecs

The engine supports and processes the following codec families:

Codec Family	Formats Included
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm_s16le, pcm_s24le, pcm_f32le, pcm_f32be
FLAC	All FLAC channel layouts

2. FFmpeg Global Options

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- `-y` — overwrites the output file without asking.
- `-loglevel error` — suppresses all output except actual errors.
- `-analyzeduration 200M / -probesize 200M` — widens the initial probe buffer to 200 MB; high-bitrate audio like TrueHD+Atmos or AAC 7.1 needs a deep probe to fully resolve the channel layout.
- `-err_detect ignore_err` — bypasses minor bitstream errors so one corrupted audio frame doesn't abort a multi-hour conversion.
- `-drc_scale 0` — disables Dynamic Range Compression, preserving the source's full dynamic range.

2.5 ScanThrottle + ThreadCount

Sets how many CPU threads the encoder uses (line 44).

```
$ThreadCount = 8      # User-adjustable (4-16).
```

Limits how many peak-scan tasks run at the same time (line 45).

```
$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount /  
$ThreadCount))  
# Recommended overrides: 1, 2, 3, or 4
```

What ScanThrottle does:

- Limits parallel SPN scans to avoid CPU overload.
- Keeps the system responsive during peak detection.
- Does not affect encoding speed.
- SSD/NVMe: safe to override up to 4. Spinning disk: keep at 1.

3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero` — clamps negative initial PTS (common in source files) to zero, preventing audio lag or lead.
- `-max_muxing_queue_size 14000` (*1024–100000 recommended range*) — raises muxing buffer headroom. Pass+Copy mode emits 3 streams (video copy, 7.1 pass, 7.1→EAC3 encode); the copy and pass streams can flood the muxer while the encode pipeline catches up, so 14000 prevents de-sync on 3+ hour files.
- `-map 0:v? -c:v copy` — maps all video tracks in stream-copy mode for zero quality loss at maximum speed.
- `-map_metadata 0 -map_chapters 0` — copies global metadata and chapter markers (title, year, scene markers) from the source.

3.2 Safe-Peak Normalizer (SPN)

- **What is:** Safe-Peak Normalizer (SPN-TP) measures each track's true peak before processing and only steps in when that true peak exceeds the ceiling (above -1.0 dBTP).
- **Why:** Some movies arrive with peaks already above 0 dBFS. When that happens, even simple mixing of the channels can push those peaks higher and cause distortion in the final encode. SPN catches this early so the issue never reaches the rest of the audio processing.
- **How:** Safe-Peak Normalizer runs after rule matching and before filters are built (Encode, Downmix, etc.). If the true-peak scan shows the track exceeds the ceiling, a static, attenuation-only trim (`volume=-X dB`) is added to that track's filter chain — after the downmix on the fold-to-5.1 paths, so the trim is computed against the exact 5.1 signal that ships. The trim only ever attenuates; it never boosts.
- **Clean Sources:** Nothing is changed — the original loudness and dynamics stay exactly the same.
- **Passthrough tracks:** — Peak detection is skipped entirely. Copied streams cannot have filters applied.
- **Dirty Sources:** SPN-TP trims unsafe true peaks back to the ceiling without making the movie sound compressed or “over-normalized.” On the two fold-to-5.1 paths the alimiter sits just below as a final sample-peak backstop.
- **Result:** More consistent, safer audio across all movies, keeping clean sources untouched.

True-Peak Ceiling (line 46)

This is the value SPN-TP uses to decide when to step in.

```
[double]$TruePeakCeilingDB = -1.0 # SPN-TP: measured true peak above this ceiling (dBTP) gets a static, attenuation-only trim back to it
```

- A track whose measured true peak exceeds this ceiling receives a static trim of $\min(0, \text{ceiling} - \text{measured_tp})$ dB — attenuation only, never a boost.
- Lower values (e.g. -2.0) trim more tracks and leave more headroom — more conservative.
- Higher values (e.g. -0.5) trim fewer tracks — only the hottest sources.
- The alimiter backstop on the two fold-to-5.1 paths sits just below at ~ -0.47 dBFS.

Three-Stage SPN Processing

The audio engine processes each audio track in three phases, with SPN peak scans running in parallel during Phase B to keep multi-track processing fast.

```
3-phase SPN architecture (Phase A > Phase B > Phase C)
```

```
Input MKV >> Probe
```

```
> Phase A (rule matching + malformed-layout guards + commentary removal)
```

```
> PreTrack-QRS (per-language winner selection – removes fallback-tier  
duplicates)
```

```
> Phase B (parallel SPN peak scans)
```

```
> Phase C (sequential merge + NeedsPeakSafety resolution)
```

```
> FFmpeg command builder
```

```
>> Output MKV
```

Phase A, Sequential rule matching

Every audio stream is probed and matched against the rule table. Malformed-layout guards and commentary detection run here. 2.0 tracks are marked Removed. Surviving tracks receive a pending action (Passthrough, Pass+Copy, Downmix, Encode). PreTrack-QRS then runs over these survivors, removing redundant fallback-tier duplicates per language before any scan work begins (see § "PreTrack-QRS" below).

Phase B, Parallel SPN peak scans

All non-Removed, non-Passthrough tracks are scanned in parallel for true-peak levels using **loudnorm** (`input_tp`). Downmix and Pass+Copy tracks are measured *through* the downmix filtergraph, so the trim matches the exact 5.1 signal that ships. DTS/DTS-HD sources with more than 6 channels (which route to Encode, not Downmix) are likewise reduced 7.1→5.1 *inside* the measure filtergraph via `aresample=ochl=5.1` before loudnorm, so their measured peak matches the same 5.1 signal the `-ac 6` encode ships.

Phase C, Sequential merge

True-peak results are merged back into the track list. **NeedsPeakSafety** and the static trim amount (`PeakSafetyGainDb`) are resolved from the scan data. The final track list is handed to the FFmpeg command builder.

3.5 PreTrack-QRS — Per-Language Winner Selection

PreTrack-QRS runs immediately after Phase A rule matching and before the Phase B scan. It deduplicates redundant low-value tracks **per language**, so the SPN scan and the encode never spend work on a fallback track that a rule-matched track already supersedes.

How selection works:

- Surviving (non-Removed) tracks are grouped by language (case-insensitive).
- Within each language group, tracks are sorted by **Priority (high → low)**, then by **original stream order** as the tiebreaker.
- The top track is the **winner** and is always kept, regardless of its priority.
- Each remaining track in the group (a "loser") is removed **only if** its priority falls below the removal threshold.

\$QRSMinPriority (line 54)

```
# Rule-matched keepers score >=61; fallbacks (plain AC-3 5.1, etc.) score 0.  
# Losers below this threshold are removed. Winner always exempt. MaxValue =  
strict last-man-standing.  
$QRSMinPriority = 60
```

- **Default 60** is deliberately conservative. Every rule-matched track carries a priority of **61 or higher** (the lowest rule-matched action is PCM/FLAC 5.1 Encode = 61), so at the default floor **no rule-matched track is ever removed** — even as a loser. Only **fallback-tier tracks** (priority 0 — e.g. a plain AC-3 5.1 that matched no explicit codec rule) are eligible for removal.
- The **winner is always exempt**, even if its priority is 0.
- Raise toward `[int]::MaxValue` for strict **last-man-standing** behavior (keep exactly one track per language, remove every other loser regardless of priority) — the reference-revert setting.
- Lowering below 60 has no additional effect: 60 already spares every rule-matched track, and fallbacks sit at 0.

Removed tracks are marked `Removed` with output `QRS_Removed` and rule `PreTrack_QRS_Removed`; they are excluded from Phase B scanning and from the final mux, exactly like 2.0/commentary removals.

Console output:

- Winner per language is printed in **green**: [PreTrack-QRS] Winner: Track N (codec, Nch, lang, profile) - Rank P [rule]
- Each removed track is printed in **red**: [PreTrack-QRS] Removed: Track N (codec, Nch, lang)

Safety guard: if a language group somehow ends with zero survivors, the script throws rather than silently shipping a language with no audio.

4. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- **-filter:a:\$i \$panFilter** — applies the 7.1→5.1 pan matrix on Downmix and Pass+Copy paths; see § 6 for the matrix and its rationale.
- **-c:a:\$i eac3 / -b:a:\$i \$t.Bitrate** — EAC3 (DD+) at the bitrate from the rule table (§ 5) or \$BitRateConfig.
- **-ac 6** — forces exactly 6 channels on the 5.1 re-encode path, guaranteeing correct channel count even on a non-standard source layout. On DTS/DTS-HD sources with more than 6 channels (Encode action), this same flag also performs the actual 7.1→5.1 reduction (swr rematrix) — these tracks are downmixed by **-ac 6**, not the pan matrix.
- **-ar 48000** — normalizes the output sample rate to 48 kHz on all EAC3 encode paths (Downmix, Pass+Copy duplicate, 5.1 re-encode). DD+ delivery is 48 kHz; this brings high-rate lossless sources (96/192 kHz PCM/FLAC) down to spec. Not applied to passthrough/copy streams.
- **-dialnorm -31** — sets Dialogue Normalization to the reference "off" position, preventing the playback device from applying its own volume leveling on top of the mix's intended loudness.
- **-cutoff 20000** — sets the low-pass filter to 20 kHz, preserving high-frequency content that some encoders discard by default. Applies to Downmix, Pass+Copy DDP duplicate, and 5.1 re-encode paths only — not passthrough.
- **-metadata:s:a:\$i "title=..."** — labels the track (e.g., "DD+ 5.1 Downmix") so it's identifiable in a media player's audio menu.

5. Audio Rules & Priority

The audio-engine uses a **Priority Mapping (0-110)** to evaluate tracks based on quality and codec type.

The "Pass+Copy" Strategy

- **Action:** Specifically for **TrueHD 7.1** and **AAC 7.1**.
- **What:** It performs a **Passthrough** of the original 7.1 master while creating a **new EAC3 (DD+) 5.1** duplicate at **1152k** (**PassCopy51**).
- **Why:** This future-proofs the file for high-end theaters while providing a high-quality fallback for standard devices that cannot decode 7.1 layouts.

- **Summary Visualization:** This action generates **two rows** in the final summary report for a single input track:
 - **Row 1:** The original 7.1 Passthrough (Master).
 - **Row 2:** The new DDP 5.1 Copy (Compatibility track).

TrueHD Rules (Priority 110)

Priority	Channels	Action	Output
110	8ch (7.1)	Pass+Copy	TrueHD 7.1 kept + DDP 5.1 @ 1152k
81	6ch (5.1)	Passthrough	Original TrueHD 5.1

EAC3 / Atmos Rules

Priority	Condition	Action	Output
100	EAC3 Atmos (JOC)	Passthrough	Original EAC3 Atmos — preserves object metadata
99	EAC3 7.1	Downmix	DDP 5.1 @ 1152k
98	EAC3 5.1	Passthrough	Original EAC3 5.1
97	EAC3 2.0	Removed	Delegated to ConvertAudioEngine-Stereo

AAC Rules

Priority	Channels	Action	Output
93	7.1 (8ch)	Pass+Copy	AAC 7.1 kept + DDP 5.1 @ 1152k
91	5.1 (6ch)	Encode	DDP 5.1 @ 768k
90	2.0 (2ch)	Removed	Delegated to ConvertAudioEngine-Stereo

Malformed AAC 7ch → corrected to 6ch before rule matching.

DTS Rules

Priority	Variant	Action	Output
73	DTS-HD MA/HRA (any ch > 2)	Encode	DDP 5.1 @ 1024k
72	DTS Core (any ch > 2)	Encode	DDP 5.1 @ 768k
71	DTS-HD 2.0	Removed	Delegated to ConvertAudioEngine-Stereo
70	DTS 2.0	Removed	Delegated to ConvertAudioEngine-Stereo

PCM / FLAC Rules

Priority	Channels	Action	Output
62	7.1 (8ch)	Downmix	DDP 5.1 @ 1152k
61	5.1 (6ch)	Encode	DDP 5.1 @ 768k
60	2.0 (2ch)	Removed	Delegated to ConvertAudioEngine-Stereo

🚫 Fallback Rules

If no specific rule matches, the following logic is applied:

Channels	Fallback Action	Bitrate
2 Channels or fewer	Removed	— (delegated to ConvertAudioEngine-Stereo)
3 to 6 channels	Encode	768k
more than 6 channels (7ch+)	Downmix	1152k

6. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

The Formula

```
aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707SL|BR=BR+0.707SR
alimiter=limit=0.948:attack=5:release=50:level=false:latency=1
```

Layout Locking (aformat)

What: Forces the decoder to output a strict 8-channel 7.1 layout *before* the Pan Filter runs.

Why: Some codecs (TrueHD, AAC, EAC3) may decode into non-standard 7.1 variants.

Locking the layout ensures:

- SL really is the Side Left channel, BL really is the Back Left channel
- No channel swapping, No silent corruption of the downmix

The Math (0.707)

What: Applies a -3 dB coefficient to the Side Surrounds (SL/SR).

Why: Prevents the side channels from overpowering the rear soundstage and avoids digital clipping.

Limiter (0.948) — Fold-to-5.1 Paths (Downmix + Pass+Copy)

alimiter=limit=0.948:attack=5:release=50:level=false:latency=1

latency=1 enables compensation for the look-ahead delay introduced by **attack**, so the limiter catches post-downmix peaks while keeping the audio in sync.

Why: Any 7.1 track that is downmixed or duplicated into 5.1 can produce true-peak overshoots after decoding:

- AAC 7.1 > when creating the DDP 5.1 copy
- TrueHD 7.1 > when creating the DDP 5.1 copy
- PCM/FLAC > during 7.1 → 5.1 downmix
- EAC3 7.1 > during 7.1 → 5.1 downmix

DTS/DTS-HD 7.1 is **not** routed through this limiter — it is reduced separately via **-ac 6** (swr), and its peaks are controlled by the static SPN-TP trim alone.

Important:

- If the source audio already contains true-peak or inter-sample peaks (“clips”), these will also appear in **analysis** of the downmix.
- The limiter prevents new clipping during downmixing and encoding, but **it does not remove overshoots inherent in the source mix**.

7. Commentary Filtering

- **Only 2-channel commentary tracks** are removed to save space while preserving primary content.
- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** commentary, director, producer, writer, cast, behind, bonus, alt, interview.
- **Why 5.1 Commentary is NOT removed:**
 - It is extremely rare in retail media.
 - Hard to tell apart from "Alternate Mixes" or "Home Theater Optimized" tracks.
 - Removing it carries a high risk of "False Positives" (deleting real movie content).

8. Malformed Layout Guards

The script corrects illegal 7-channel layouts before rule matching runs:

- **AAC 7ch** → 6ch (treated as 5.1)
- **EAC3 7ch** → 6ch (treated as 5.1)
- **TrueHD 7ch** → 8ch (treated as 7.1)
- **PCM/FLAC 7ch** → 8ch (treated as 7.1)
- **AC3 7ch** → Removed (no valid AC3 7-channel layout exists)

9. Audio Processing Summary

- For Pass+Copy tracks, \ - - Copy appears as Action and output identifies it as DDP5.1 audio.

Column	Description
Index	The original stream number (0, 1, 2).
Codec	The source format (e.g., TrueHD, DTS).
Channels	The detected number of channels (e.g., 7.1, 5.1).
Action	Green : Passthrough / Pass, Yellow : Downmix / Copy, Blue : Encode, Red : Removed.
Output	The final codec and bitrate configuration.
Priority	The assigned priority level (0–110). Removed tracks show —.
Rule	The logical rule that was matched during processing.

Subtitles and attachments are always carried through unchanged. The script maps `-map 0:s?` (all subtitles) and `-map 0:t?` (all attachments — fonts, cover art) with stream copy. There is no toggle to disable this.

10. 2.0 Audio Encoding — Why It's Disabled

All 2.0 rules are present in the engine but set to `Action = "Removed"` by default. This is intentional — stereo output is delegated to **ConvertAudioEngine-Stereo**, and **the engine has no native stereo encode path**. That second point is what makes re-enabling a two-step change, not a one-step one.

⚠ **Changing the rule alone upmixes 2.0 → 5.1.** Any `Action = "Encode"` track is routed through the 5.1 re-encode branch of `Build-FFmpegCommand`, which unconditionally applies `-ac 6`. Flip a 2.0 rule to `Encode` and nothing else, and the source is **upmixed to DD+ 5.1** at your chosen bitrate — *not* emitted as a 2.0 track. For most uses that is not what you want; leaving 2.0 delegated to `ConvertAudioEngine-Stereo` is the recommended default.

11. Non-Critical FFmpeg Warnings (**Safe to Ignore**)

During processing, FFmpeg may display one or more of the following messages.

These warnings are **normal**, and do not affect audio quality, sync, or re-encode.

They come from FFmpeg's internal decoders and subtitle parsers — not from the script.

“quant_step_size larger than huff_lsbs”

- **Cause:** Printed by the TrueHD decoder during the initial probe phase.
- **Impact:** Safe to ignore. Does not affect decoding, channel layout, or the Pass+Copy path.

“Codec AVOption drc_scale ... has not been used for any stream”

- **Cause:** (`-drc_scale`) applies only to AC-3/EAC3 decoders.
FFmpeg prints this when the input track is not AC-3 (e.g., TrueHD, DTS, FLAC).
- **Impact:** Safe to ignore. The script disables DRC intentionally for consistent output.

“(Subtitle: hdmv_pgs_subtitle) unspecified size”

- **Cause:** PGS (Blu-ray) subtitles do not declare width/height in the container.
FFmpeg warns until the first subtitle packet is decoded.
- **Impact:** Safe to ignore. Subtitles are copied untouched (`-c:s copy`).

“Assuming an incorrectly encoded 7.1 channel layout...instead of 7.1(wide)”

- **Cause:** AAC has two 7.1 layouts. Most encoders use the common (non-spec) layout.
FFmpeg prints this when the file does not use the strict 7.1(wide) AAC layout.
- **Impact:** Safe to ignore. Script corrects and pins layout using (`aformat=channel_layouts=7.1`)

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.